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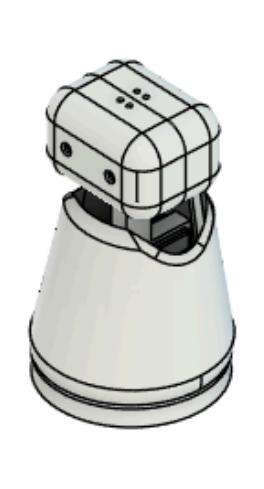
Study Report



Analyzed File	0 - Assembly v7	Version	Autodesk Fusion (2702.1.58)
Creation Date	2026-05-27, 07:26:38	Author	User

Report Properties

Report Properties	
Title	Studies
Author	User

Simulation Model 1

Study 3 - Static Stress

Study Properties

Study Properties	
Study Type	Static Stress
Last Modification Date	2026-05-27, 07:15:11

Study Setup

Settings	
General	
Contact Tolerance	0.10 mm
Remove Rigid Body Modes	No
Mesh	
Average Element Size (% of model size) - Solids	10
Scale Mesh Size Per Part	No
Average Element Size (absolute value)	-
Element Order	Parabolic
Create Curved Mesh Elements	No
Max. Turn Angle on Curves (Deg.)	60
Max. Adjacent Mesh Size Ratio	1.5
Max. Aspect Ratio	10
Minimum Element Size (% of average size)	20
Adaptive Mesh Refinement	
Number of Refinement Steps	0
Results Convergence Tolerance (%)	20
Portion of Elements to Refine (%)	10
Results for Baseline Accuracy	von Mises Stress

Materials	
Stainless Steel	
Components	
Component	Strength
2 - Adapter Assem v2:1/2 - Base Mech v2:1	Yield Strength
2 - Adapter Assem v2:1/flange coupling 5mm v1:1	Yield Strength
Nema17 v2:1/Base part:1	Yield Strength
Nema17 v2:1/Top part:1	Yield Strength
Nema17 v2:1/body:1	Yield Strength
Nema17 v2:1/shaft:1	Yield Strength
Nema17 v2:1/supply one:1	Yield Strength
Properties	
Density	8.000E-06 kg / mm^3
Young's Modulus	193000.00 MPa
Poisson's Ratio	0.30
Yield Strength	250.00 MPa
Ultimate Tensile Strength	540.00 MPa
Thermal Conductivity	0.016 W / (mm C)
Thermal Expansion Coefficient	1.040E-05 / C
Specific Heat	477.00 J / (kg C)
ABS Plastic	
Components	
Component	Strength
1 - Base 3D v3:1	Yield Strength
2 - Base SS v3:1	Yield Strength
3 - Body v17:1	Yield Strength
4 - Front Assem v3:1/4 - Head Front v6:1	Yield Strength
4 - Head v27:1	Yield Strength
5 - mech0 v8:1/5 - mech1 v6:1	Yield Strength
5 - mech0 v8:1/5 - mech2 v3:1	Yield Strength
5 - mech0 v8:1/5 - mech3 v8:1	Yield Strength
5 - mech0 v8:1/5 - mech4 v17:1	Yield Strength
5 - mech0 v8:1/5 - mech5 v6:1	Yield Strength
Properties	
Density	1.060E-06 kg / mm^3
Young's Modulus	2240.00 MPa
Poisson's Ratio	0.38
Yield Strength	20.00 MPa
Ultimate Tensile Strength	29.60 MPa
Thermal Conductivity	1.600E-04 W / (mm C)
Thermal Expansion Coefficient	8.570E-05 / C
Specific Heat	1500.00 J / (kg C)
Plastic, Opaque Black	
Components	
Component	Strength
5 - mech0 v8:1/mg996R_v2 v1:1/Body1	Yield Strength
5 - mech0 v8:1/mg996R_v2 v1:1/Body2	Yield Strength
5 - mech0 v8:1/mg996R_v2 v1:1/Body3	Yield Strength
5 - mech0 v8:1/mg996R_v2 v1:1/Body4	Yield Strength
5 - mech0 v8:1/mg996R_v2 v1:1/Body5	Yield Strength
5 - mech0 v8:1/mg996R_v2 v1:1/Body6	Yield Strength
5 - mech0 v8:1/mg996R_v2 v1:1/Body7	Yield Strength
5 - mech0 v8:1/mg996R_v2 v1:1/Body8	Yield Strength
5 - mech0 v8:1/mg996R_v2 v1:2/Body1	Yield Strength
5 - mech0 v8:1/mg996R_v2 v1:2/Body2	Yield Strength
5 - mech0 v8:1/mg996R_v2 v1:2/Body3	Yield Strength
5 - mech0 v8:1/mg996R_v2 v1:2/Body4	Yield Strength
5 - mech0 v8:1/mg996R_v2 v1:2/Body5	Yield Strength
5 - mech0 v8:1/mg996R_v2 v1:2/Body6	Yield Strength
5 - mech0 v8:1/mg996R_v2 v1:2/Body7	Yield Strength
5 - mech0 v8:1/mg996R_v2 v1:2/Body8	Yield Strength
5 - mech0 v8:1/mg996_arm_2pad v2:1	Yield Strength
5 - mech0 v8:1/mg996_arm_2pad v2:2	Yield Strength
Properties	
Density	9.520E-07 kg / mm^3
Young's Modulus	911.00 MPa
Poisson's Ratio	0.392
Yield Strength	20.67 MPa
Ultimate Tensile Strength	20.67 MPa
Thermal Conductivity	2.110E-04 W / (mm C)
Thermal Expansion Coefficient	1.500E-04 / C
Specific Heat	2859.00 J / (kg C)

Contacts

Contact Type	Contact Set	Penetration Type	Bodies	Entities
Bonded				
	Bonded1	Symmetric	[Body1 flange coupling 5mm v1:1||Body1 2 - Base Mech v2:1]	[Face 107||Face 106]
	Bonded2	Symmetric	[Body5 4 - Head Front v6:1||Body5 4 - Head v27:1]	[Face 475||Face 365]
	Bonded3	Symmetric	[Body5 4 - Head Front v6:1||Body5 4 - Head v27:1]	[Face 474||Face 360]
	Bonded4	Symmetric	[Body5 4 - Head Front v6:1||Body5 4 - Head v27:1]	[Face 473||Face 359]
	Bonded5	Symmetric	[Body5 4 - Head Front v6:1||Body5 4 - Head v27:1]	[Face 472||Face 358]
	Bonded6	Symmetric	[Body5 4 - Head Front v6:1||Body5 4 - Head v27:1]	[Face 471||Face 357]
	Bonded7	Symmetric	[Body5 4 - Head Front v6:1||Body5 4 - Head v27:1]	[Face 470||Face 364]
	Bonded8	Symmetric	[Body5 4 - Head Front v6:1||Body5 4 - Head v27:1]	[Face 469||Face 363]
	Bonded9	Symmetric	[Body5 4 - Head Front v6:1||Body5 4 - Head v27:1]	[Face 468||Face 362]
	Bonded10	Symmetric	[Body5 4 - Head Front v6:1||Body5 4 - Head v27:1]	[Face 467||Face 361]
	Bonded11	Symmetric	[Body1 mg996_arm_2pad v2:2||Body1 5 - mech4 v17:1]	[Face 1577||Face 82]
	Bonded12	Symmetric	[Body1 mg996_arm_2pad v2:1||Body1 5 - mech5 v6:1]	[Face 1577||Face 82]
	Bonded13	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8838||Face 1482]
	Bonded14	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8796||Face 1515]
	Bonded15	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8783||Face 1503]
	Bonded16	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1531]
	Bonded17	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1528]
	Bonded18	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1525]
	Bonded19	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1522]
	Bonded20	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1519]
	Bonded21	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1516]
	Bonded22	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1513]
	Bonded23	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1510]
	Bonded24	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1507]
	Bonded25	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1504]
	Bonded26	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1501]
	Bonded27	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1498]
	Bonded28	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1495]
	Bonded29	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1492]
	Bonded30	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1489]
	Bonded31	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1486]
	Bonded32	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1483]
	Bonded33	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1480]
	Bonded34	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1477]
	Bonded35	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1474]
	Bonded36	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1471]
	Bonded37	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1468]
	Bonded38	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1465]
	Bonded39	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1462]
	Bonded40	Symmetric	[Body2 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8781||Face 1459]
	Bonded41	Symmetric	[Body1 mg996R_v2 v1:2||Body2 mg996R_v2 v1:2]	[Face 8518||Face 8858]
	Bonded42	Symmetric	[Body1 mg996R_v2 v1:2||Body2 mg996R_v2 v1:2]	[Face 8517||Face 8781]
	Bonded43	Symmetric	[Body1 mg996R_v2 v1:2||Body1 mg996_arm_2pad v2:2]	[Face 8516||Face 1458]
	Bonded44	Symmetric	[Body1 mg996R_v2 v1:2||Body3 mg996R_v2 v1:2]	[Face 8511||Face 9266]
	Bonded45	Symmetric	[Body1 mg996R_v2 v1:2||Body3 mg996R_v2 v1:2]	[Face 8510||Face 9265]
	Bonded46	Symmetric	[Body1 mg996R_v2 v1:2||Body3 mg996R_v2 v1:2]	[Face 8509||Face 9267]
	Bonded47	Symmetric	[Body1 mg996R_v2 v1:2||Body3 mg996R_v2 v1:2]	[Face 8508||Face 9270]
	Bonded48	Symmetric	[Body1 mg996R_v2 v1:2||Body5 mg996R_v2 v1:2]	[Face 8479||Face 10908]
	Bonded49	Symmetric	[Body1 mg996R_v2 v1:2||Body7 mg996R_v2 v1:2]	[Face 8478||Face 11094]
	Bonded50	Symmetric	[Body1 mg996R_v2 v1:2||Body5 mg996R_v2 v1:2]	[Face 8452||Face 10907]
	Bonded51	Symmetric	[Body1 mg996R_v2 v1:2||Body4 mg996R_v2 v1:2]	[Face 8451||Face 10815]
	Bonded52	Symmetric	[Body1 mg996R_v2 v1:2||Body6 mg996R_v2 v1:2]	[Face 8450||Face 11001]
	Bonded53	Symmetric	[Body1 mg996R_v2 v1:2||Body7 mg996R_v2 v1:2]	[Face 8449||Face 11093]
	Bonded54	Symmetric	[Body1 mg996R_v2 v1:2||Body5 mg996R_v2 v1:2]	[Face 8435||Face 10905]
	Bonded55	Symmetric	[Body1 mg996R_v2 v1:2||Body4 mg996R_v2 v1:2]	[Face 8434||Face 10814]
	Bonded56	Symmetric	[Body1 mg996R_v2 v1:2||Body6 mg996R_v2 v1:2]	[Face 8432||Face 11000]
	Bonded57	Symmetric	[Body1 mg996R_v2 v1:2||Body7 mg996R_v2 v1:2]	[Face 8429||Face 11091]
	Bonded58	Symmetric	[Body1 mg996R_v2 v1:2||Body8 mg996R_v2 v1:2]	[Face 8428||Face 11164]
	Bonded59	Symmetric	[Body1 mg996R_v2 v1:2||Body4 mg996R_v2 v1:2]	[Face 8423||Face 10812]
	Bonded60	Symmetric	[Body1 mg996R_v2 v1:2||Body6 mg996R_v2 v1:2]	[Face 8419||Face 10998]
	Bonded61	Symmetric	[Body1 mg996R_v2 v1:2||Body8 mg996R_v2 v1:2]	[Face 8416||Face 11164]
	Bonded62	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8824||Face 1466]
	Bonded63	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8821||Face 1463]
	Bonded64	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8818||Face 1460]
	Bonded65	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8815||Face 1532]
	Bonded66	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8800||Face 1517]
	Bonded67	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8797||Face 1514]
	Bonded68	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8785||Face 1502]
	Bonded69	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1531]
	Bonded70	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1528]
	Bonded71	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1525]
	Bonded72	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1522]
	Bonded73	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1519]
	Bonded74	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1516]
	Bonded75	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1513]
	Bonded76	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1510]
	Bonded77	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1507]
	Bonded78	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1504]
	Bonded79	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1501]
	Bonded80	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1498]
	Bonded81	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1495]
	Bonded82	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1492]
	Bonded83	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1489]
	Bonded84	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1486]
	Bonded85	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1483]
	Bonded86	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1480]
	Bonded87	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1477]
	Bonded88	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1474]
	Bonded89	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1471]
	Bonded90	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1468]
	Bonded91	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1465]
	Bonded92	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1462]
	Bonded93	Symmetric	[Body2 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8781||Face 1459]
	Bonded94	Symmetric	[Body1 mg996R_v2 v1:1||Body2 mg996R_v2 v1:1]	[Face 8518||Face 8858]
	Bonded95	Symmetric	[Body1 mg996R_v2 v1:1||Body2 mg996R_v2 v1:1]	[Face 8517||Face 8781]
	Bonded96	Symmetric	[Body1 mg996R_v2 v1:1||Body1 mg996_arm_2pad v2:1]	[Face 8516||Face 1458]
	Bonded97	Symmetric	[Body1 mg996R_v2 v1:1||Body3 mg996R_v2 v1:1]	[Face 8511||Face 9266]
	Bonded98	Symmetric	[Body1 mg996R_v2 v1:1||Body3 mg996R_v2 v1:1]	[Face 8510||Face 9265]
	Bonded99	Symmetric	[Body1 mg996R_v2 v1:1||Body3 mg996R_v2 v1:1]	[Face 8509||Face 9267]
	Bonded100	Symmetric	[Body1 mg996R_v2 v1:1||Body3 mg996R_v2 v1:1]	[Face 8508||Face 9270]
	Bonded101	Symmetric	[Body1 mg996R_v2 v1:1||Body5 mg996R_v2 v1:1]	[Face 8479||Face 10908]
	Bonded102	Symmetric	[Body1 mg996R_v2 v1:1||Body7 mg996R_v2 v1:1]	[Face 8478||Face 11094]
	Bonded103	Symmetric	[Body1 mg996R_v2 v1:1||Body5 mg996R_v2 v1:1]	[Face 8452||Face 10907]
	Bonded104	Symmetric	[Body1 mg996R_v2 v1:1||Body4 mg996R_v2 v1:1]	[Face 8451||Face 10815]
	Bonded105	Symmetric	[Body1 mg996R_v2 v1:1||Body6 mg996R_v2 v1:1]	[Face 8450||Face 11001]
	Bonded106	Symmetric	[Body1 mg996R_v2 v1:1||Body7 mg996R_v2 v1:1]	[Face 8449||Face 11093]
	Bonded107	Symmetric	[Body1 mg996R_v2 v1:1||Body1 mg996R_v2 v1:2]	[Face 8436||Face 8412]
	Bonded108	Symmetric	[Body1 mg996R_v2 v1:1||Body5 mg996R_v2 v1:1]	[Face 8435||Face 10905]
	Bonded109	Symmetric	[Body1 mg996R_v2 v1:1||Body4 mg996R_v2 v1:1]	[Face 8434||Face 10814]
	Bonded110	Symmetric	[Body1 mg996R_v2 v1:1||Body6 mg996R_v2 v1:1]	[Face 8432||Face 11000]
	Bonded111	Symmetric	[Body1 mg996R_v2 v1:1||Body7 mg996R_v2 v1:1]	[Face 8429||Face 11091]
	Bonded112	Symmetric	[Body1 mg996R_v2 v1:1||Body8 mg996R_v2 v1:1]	[Face 8428||Face 11164]
	Bonded113	Symmetric	[Body1 mg996R_v2 v1:1||Body4 mg996R_v2 v1:1]	[Face 8423||Face 10812]
	Bonded114	Symmetric	[Body1 mg996R_v2 v1:1||Body6 mg996R_v2 v1:1]	[Face 8419||Face 10998]
	Bonded115	Symmetric	[Body1 mg996R_v2 v1:1||Body1 mg996R_v2 v1:2]	[Face 8418||Face 8418]
	Bonded116	Symmetric	[Body1 mg996R_v2 v1:1||Body8 mg996R_v2 v1:1]	[Face 8416||Face 11164]
	Bonded117	Symmetric	[Body1 mg996R_v2 v1:1||Body1 mg996R_v2 v1:2]	[Face 8412||Face 8436]
	Bonded118	Symmetric	[Body3 5 - mech1 v6:1||Body5 4 - Head Front v6:1]	[Face 268||Face 486]
	Bonded119	Symmetric	[Body3 5 - mech1 v6:1||Body5 4 - Head Front v6:1]	[Face 267||Face 487]
	Bonded120	Symmetric	[Body3 5 - mech1 v6:1||Body5 4 - Head v27:1]	[Face 266||Face 367]
	Bonded121	Symmetric	[Body3 5 - mech1 v6:1||Body5 4 - Head Front v6:1]	[Face 266||Face 478]
	Bonded122	Symmetric	[Body3 5 - mech1 v6:1||Body1 5 - mech5 v6:1]	[Face 260||Face 86]
	Bonded123	Symmetric	[Body3 5 - mech1 v6:1||Body1 5 - mech4 v17:1]	[Face 259||Face 86]
	Bonded124	Symmetric	[Body1 5 - mech3 v8:1||Body8 mg996R_v2 v1:2]	[Face 419||Face 11170]
	Bonded125	Symmetric	[Body1 5 - mech3 v8:1||Body1 mg996R_v2 v1:2]	[Face 411||Face 8437]
	Bonded126	Symmetric	[Body1 5 - mech3 v8:1||Body8 mg996R_v2 v1:1]	[Face 411||Face 11170]
	Bonded127	Symmetric	[Body1 5 - mech3 v8:1||Body1 mg996R_v2 v1:2]	[Face 410||Face 8426]
	Bonded128	Symmetric	[Body1 5 - mech3 v8:1||Body1 3 - Body v17:1]	[Face 380||Face 310]
	Bonded129	Symmetric	[Body1 5 - mech3 v8:1||Body1 2 - Base Mech v2:1]	[Face 378||Face 105]
	Bonded130	Symmetric	[Body1 5 - mech2 v3:1||Body3 5 - mech1 v6:1]	[Face 149||Face 252]
	Bonded131	Symmetric	[Body1 5 - mech2 v3:1||Body5 4 - Head Front v6:1]	[Face 147||Face 487]
	Bonded132	Symmetric	[Body1 5 - mech2 v3:1||Body3 5 - mech1 v6:1]	[Face 147||Face 269]
	Bonded133	Symmetric	[Body1 5 - mech2 v3:1||Body1 5 - mech3 v8:1]	[Face 146||Face 418]
	Bonded134	Symmetric	[Body1 5 - mech2 v3:1||Body1 5 - mech3 v8:1]	[Face 144||Face 419]
	Bonded135	Symmetric	[Body1 shaft:1||Body1 flange coupling 5mm v1:1]	[Face 100||Face 105]
	Bonded136	Symmetric	[Body1 body:1||Body1 supply one:1]	[Face 348||Face 724]
	Bonded137	Symmetric	[Body1 Top part:1||Body1 shaft:1]	[Face 2141||Face 100]
	Bonded138	Symmetric	[Body1 Top part:1||Body1 body:1]	[Face 2136||Face 347]
	Bonded139	Symmetric	[Body1 Top part:1||Body1 body:1]	[Face 2135||Face 351]
	Bonded140	Symmetric	[Body1 Top part:1||Body1 body:1]	[Face 2134||Face 353]
	Bonded141	Symmetric	[Body1 Top part:1||Body1 body:1]	[Face 2133||Face 355]
	Bonded142	Symmetric	[Body1 Top part:1||Body1 body:1]	[Face 2132||Face 349]
	Bonded143	Symmetric	[Body1 Base part:1||Body1 shaft:1]	[Face 460||Face 90]
	Bonded144	Symmetric	[Body1 Base part:1||Body1 shaft:1]	[Face 457||Face 93]
	Bonded145	Symmetric	[Body1 Base part:1||Body1 body:1]	[Face 457||Face 346]
	Bonded146	Symmetric	[Body1 Base part:1||Body1 body:1]	[Face 456||Face 349]
	Bonded147	Symmetric	[Body1 Base part:1||Body1 body:1]	[Face 455||Face 355]
	Bonded148	Symmetric	[Body1 Base part:1||Body1 body:1]	[Face 454||Face 353]
	Bonded149	Symmetric	[Body1 Base part:1||Body1 body:1]	[Face 453||Face 351]
	Bonded150	Symmetric	[Body2 2 - Base SS v3:1||Body1 Base part:1]	[Face 67||Face 452]
	Bonded151	Symmetric	[Body1 1 - Base 3D v3:1||Body2 2 - Base SS v3:1]	[Face 129||Face 68]

Mesh	
Type	Solids
Nodes	460831
Elements	279359

Load Case1

Constraints

Fixed1

Type	Fixed
Ux	Fixed
Uy	Fixed
Uz	Fixed

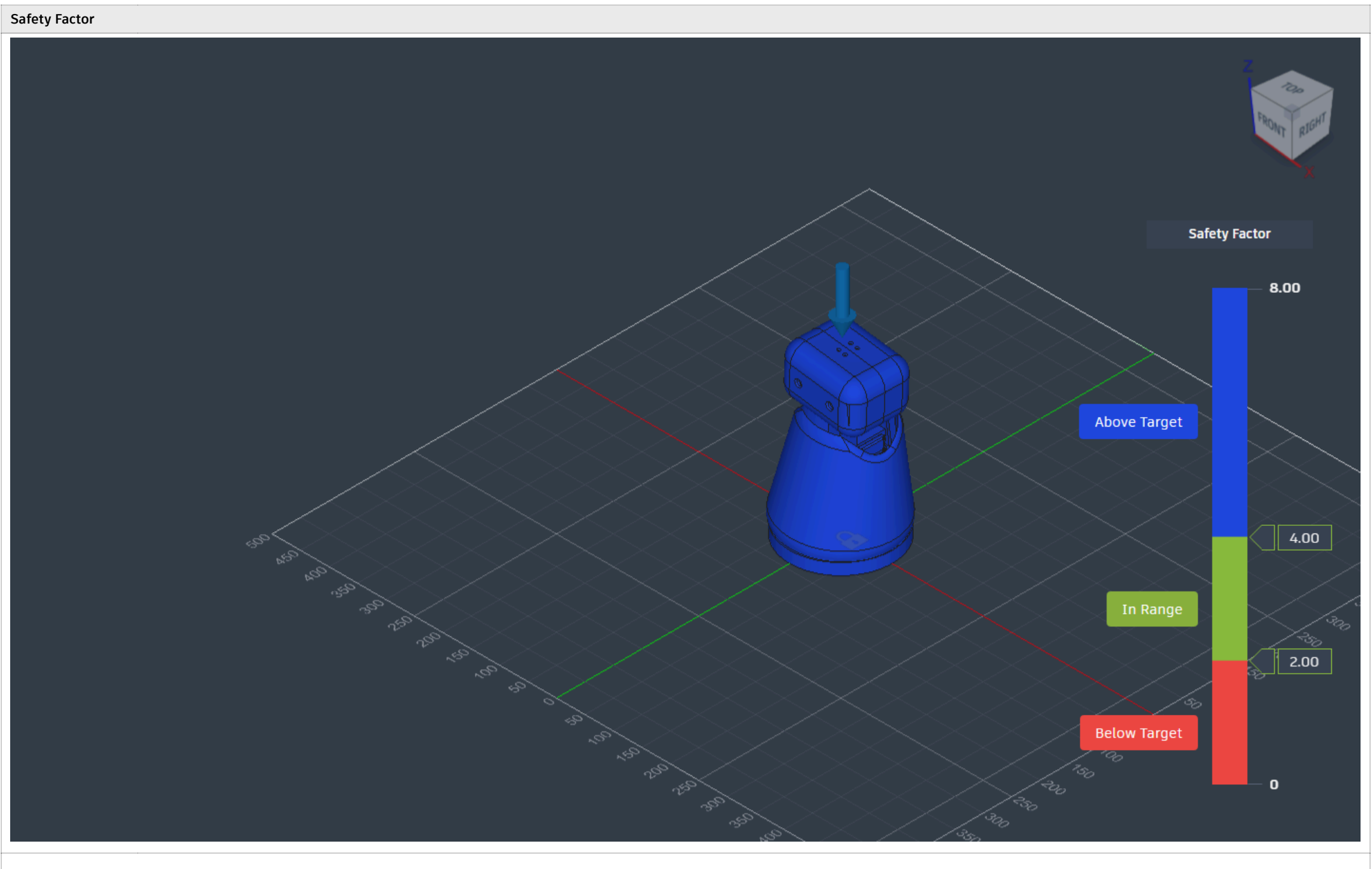
Loads

Force1

Type	Force
X Value	0.00 N
Y Value	0.00 N
Z Value	-5.00 N
Force Per Entity	No

Load Case1 - Guided Results

Safety Factor



With the analysis criteria in this study setup, the design is expected to bend permanently or break.

Next Steps

Below Safety Factor Target

Validate your study setup
- Check the loads are not set too high
- Check the mesh refinement and consider increasing it in areas of high stress
- Confirm your Safety Factor requirement
- If the Displacement result shows high displacement, consider running a Nonlinear study to gain more insight

Check other results
- Examine the Below target Result visualization to see where the weakest areas are
- Check the stress result to see if, and where, the material might yield
- Check the displacement result to see how far from its original alignment the model has moved
- Look at other results, such as reaction forces to check your study set up, or strain to check your material performance

Strengthen weak areas
- Add more material to weak areas
- Select a material with a higher yield strength
- Stiffen the design
- Modify the shape with a Generative Design study

Generate alternative designs
- Add more material to weak areas
- Select a material with a higher yield strength
- Stiffen the design
- Modify the shape with a Generative Design study

Above Safety Factor Limit

Validate your study setup
- Check the loads are not set too high
- Check the mesh refinement and consider increasing it in areas of high stress
- Confirm your Safety Factor requirement
- If the Displacement result shows high displacement, consider running a Nonlinear study to gain more insight

Check other results
- Examine the Below target Result visualization to see where the weakest areas are
- Check the stress result to see if, and where, the material might yield
- Check the displacement result to see how far from its original alignment the model has moved
- Look at other results, such as reaction forces to check your study set up, or strain to check your material performance

Lightweight the design
- Set a Safety Factor upper limit to find where material can be removed
- Remove material from areas that exceed the Safety Factor upper limit

Generate alternative designs
- In the Generative Design workspace, create design modifications
- Remove unnecessary material using a Shape Optimization study

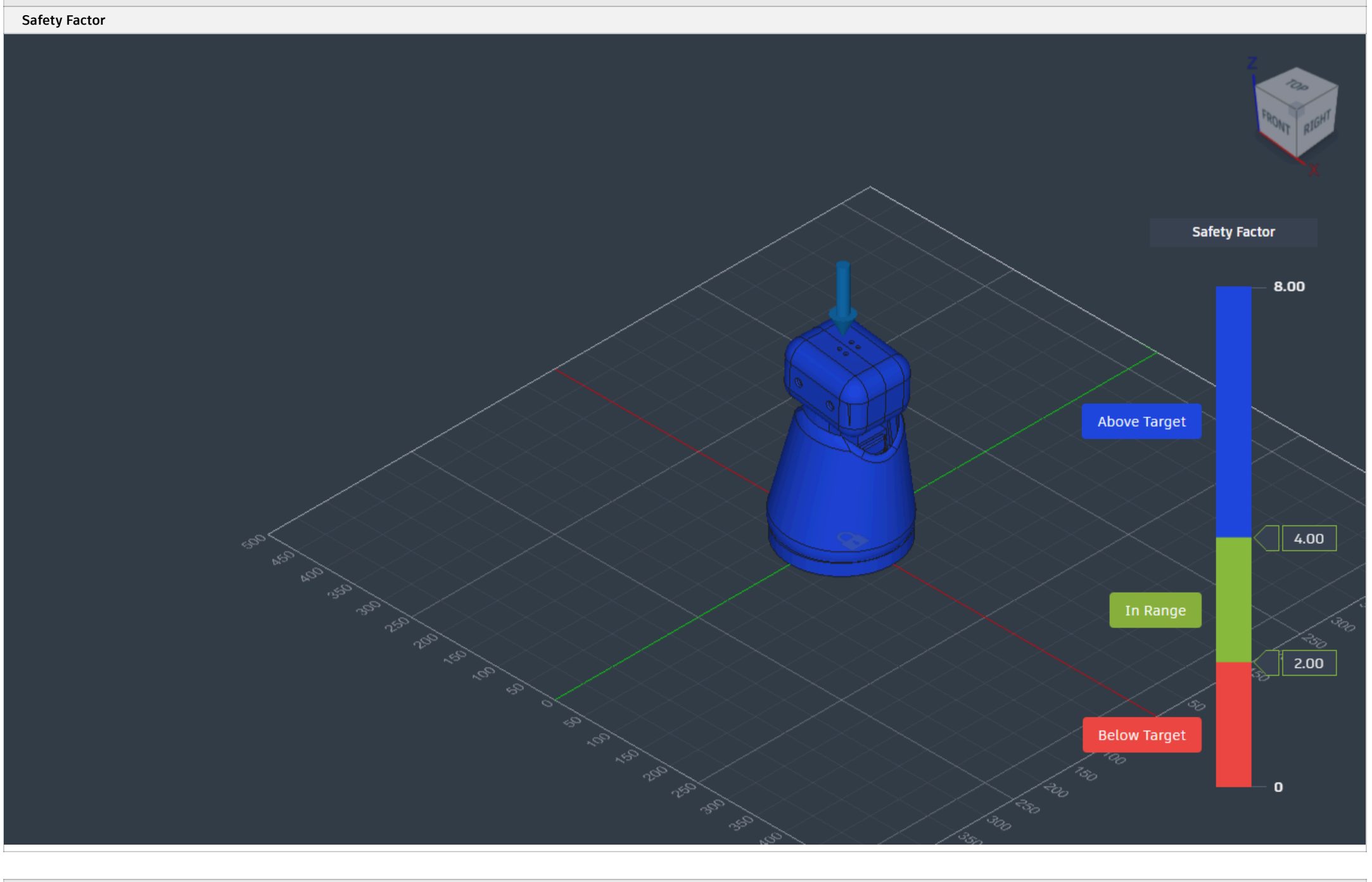
Load Case1 - Results

Result Summary

Result type	Sub-result	Minimum	Maximum
Safety Factor			
	Safety Factor	151.673	8.085E+09
Stress			
	Von Mises	2.557E-09 MPa	1.51 MPa
	1st Principal	-0.307 MPa	1.486 MPa
	3rd Principal	-1.517 MPa	0.602 MPa
	Normal XX	-1.176 MPa	1.387 MPa
	Normal YY	-1.133 MPa	1.32 MPa
	Normal ZZ	-1.517 MPa	0.999 MPa
	Shear XY	-0.568 MPa	0.555 MPa
	Shear YZ	-0.291 MPa	0.31 MPa
	Shear XZ	-0.298 MPa	0.252 MPa
Displacement			
	Total	0 mm	0.021 mm
	X	-0.002 mm	0.016 mm
	Y	-0.006 mm	7.288E-04 mm
	Z	-0.013 mm	0.003 mm
Reaction Force			
	Total	0 N	0.021 N
	X	-0.008 N	0.009 N
	Y	-0.007 N	0.008 N
	Z	-0.001 N	0.021 N
Strain			
	Equivalent	4.995E-13	1.57E-04
	1st Principal	-1.111E-08	1.267E-04
	3rd Principal	-1.651E-04	8.438E-08
	Normal XX	-3.705E-05	3.48E-05
	Normal YY	-1.162E-04	5.916E-05
	Normal ZZ	-8.148E-05	3.396E-05
	Shear XY	-1.097E-04	6.661E-05
	Shear YZ	-9.204E-05	6.161E-05
	Shear XZ	-1.023E-04	8.129E-05
Contact Pressure			
	Total	0 MPa	0.995 MPa
	X	-0.718 MPa	0.785 MPa
	Y	-0.977 MPa	0.701 MPa
	Z	-0.333 MPa	0.721 MPa
Contact Force			
	Total	0 N	2.724 N
	X	-1.313 N	1.575 N
	Y	-1.351 N	1.484 N
	Z	-2.557 N	1.756 N

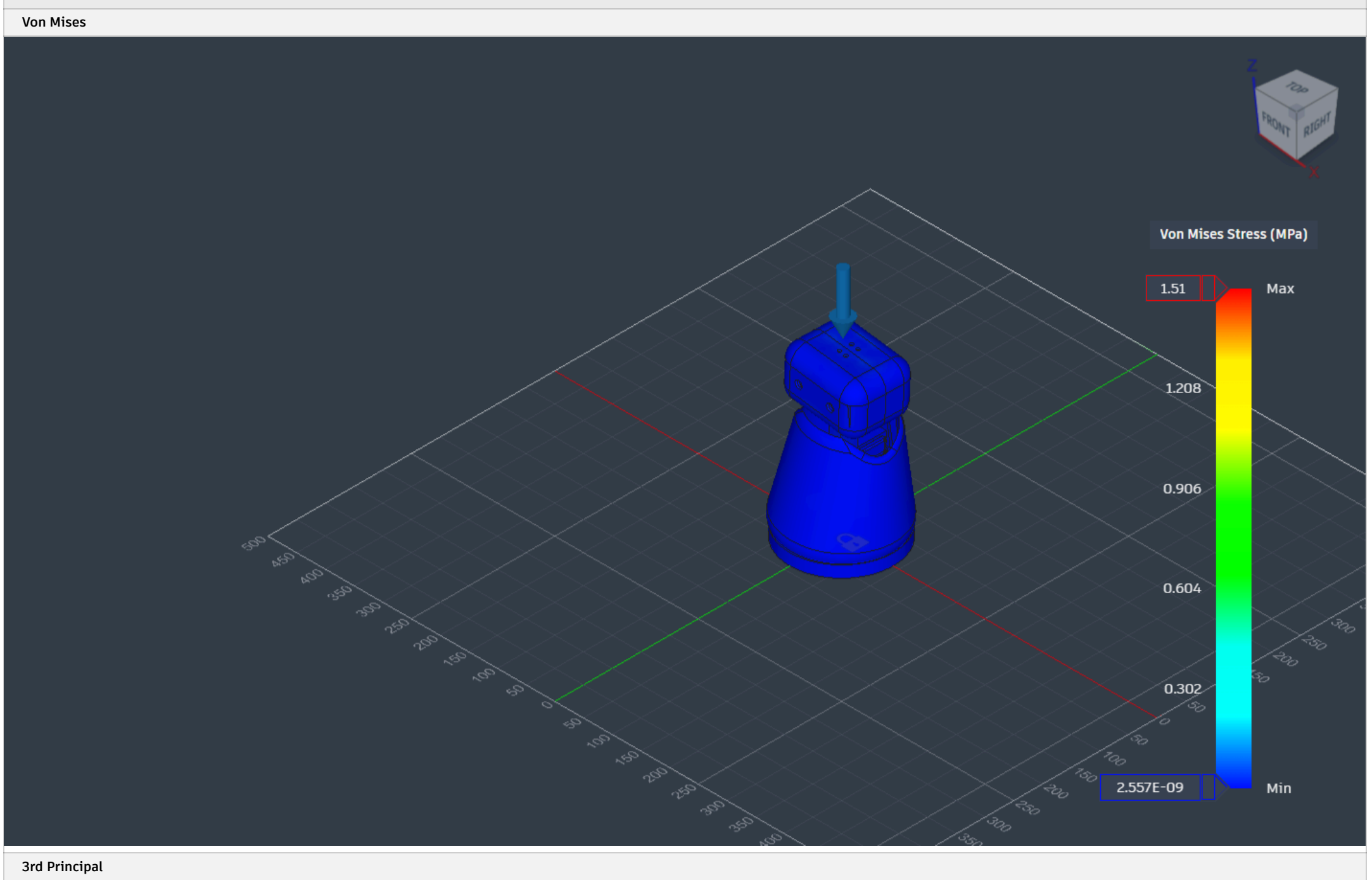
Safety Factor

Safety Factor

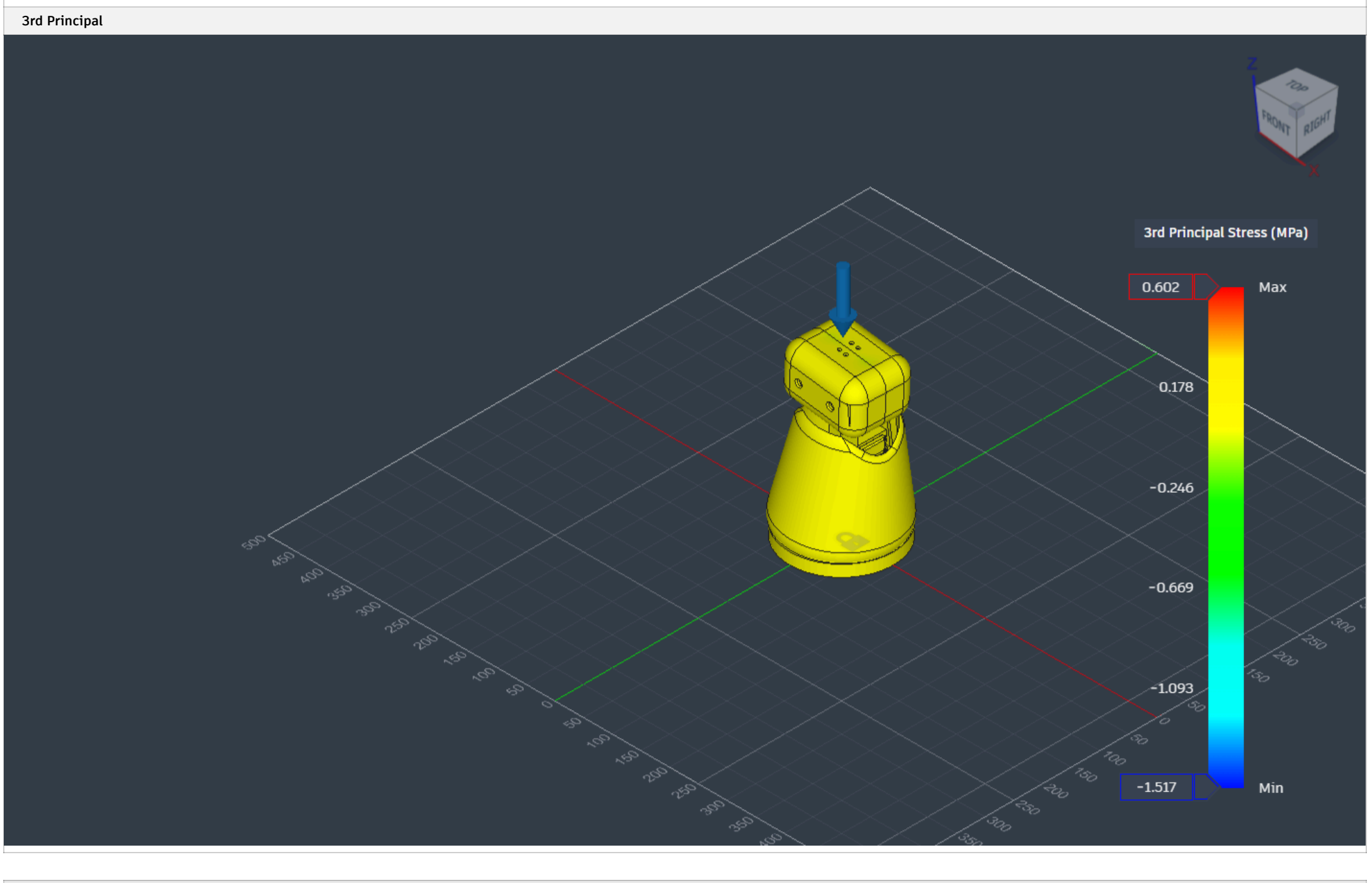


Stress

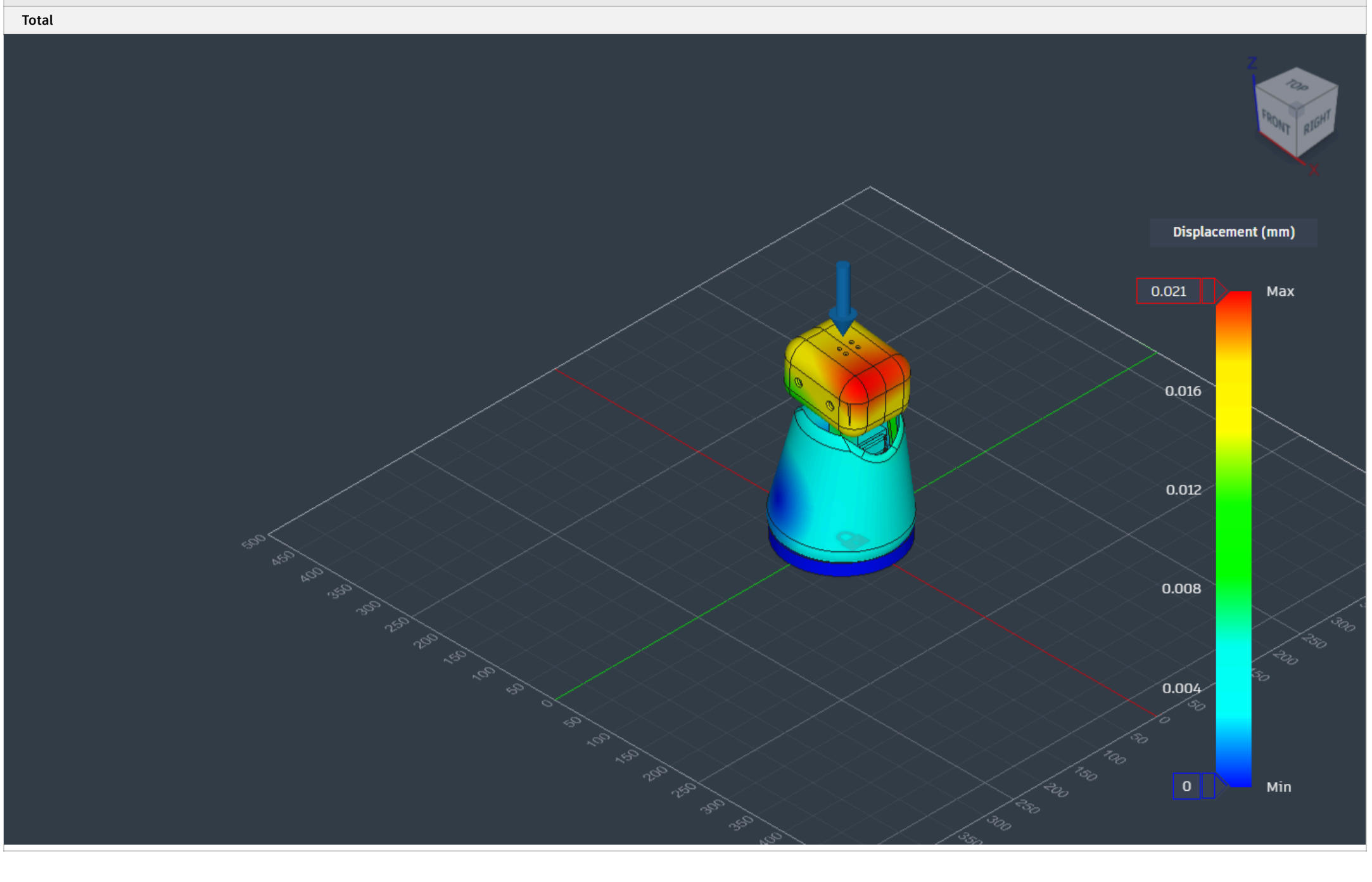
Von Mises



3rd Principal



Displacement

Total